ECEN/CSEN 123 Lab 2 Report

Tuesday 2:15 - 5:00

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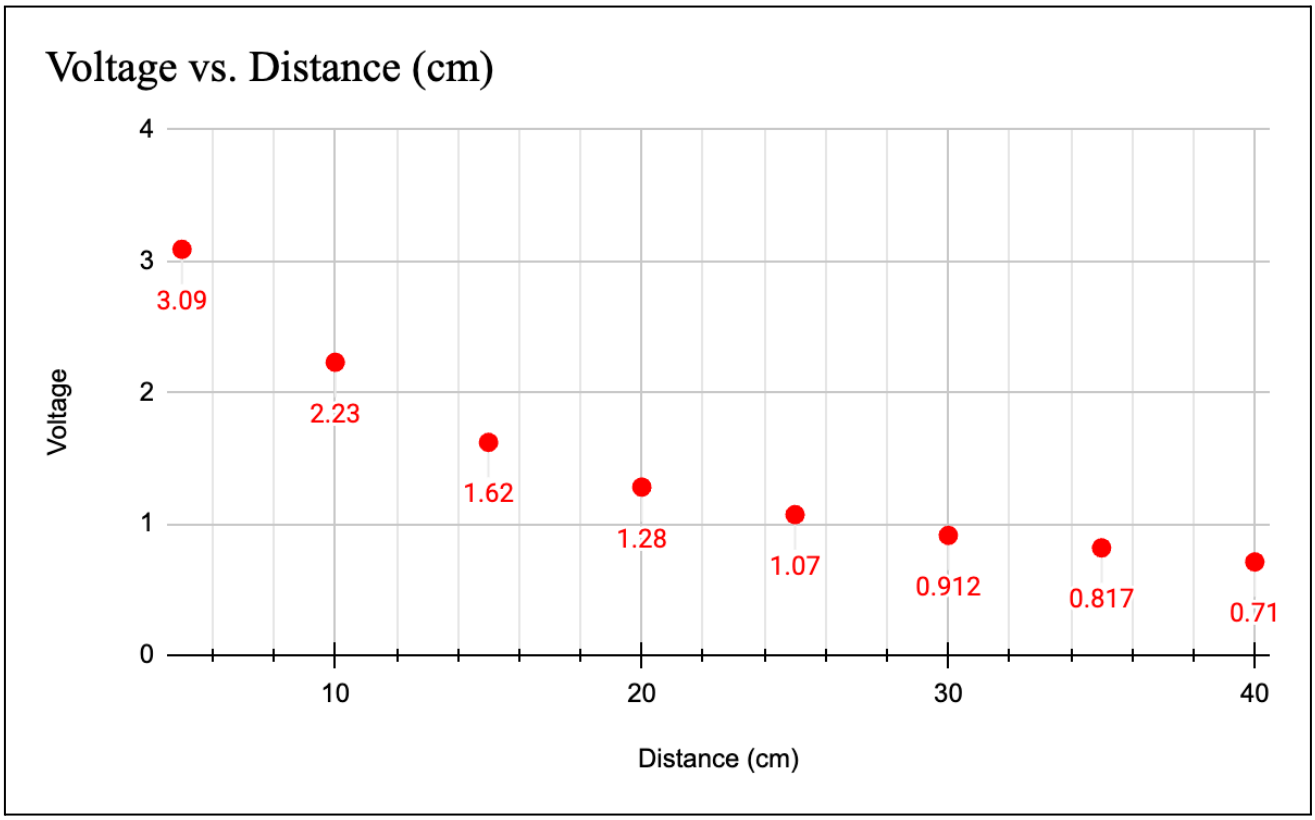
I. OBJECTIVES

- To learn how to operate and characterize a sensor.
- To learn how to construct a sensor
- To learn how to design a simple amplifier circuit

Part I: Operating and characterizing the Sharp GP2Y0A21YK

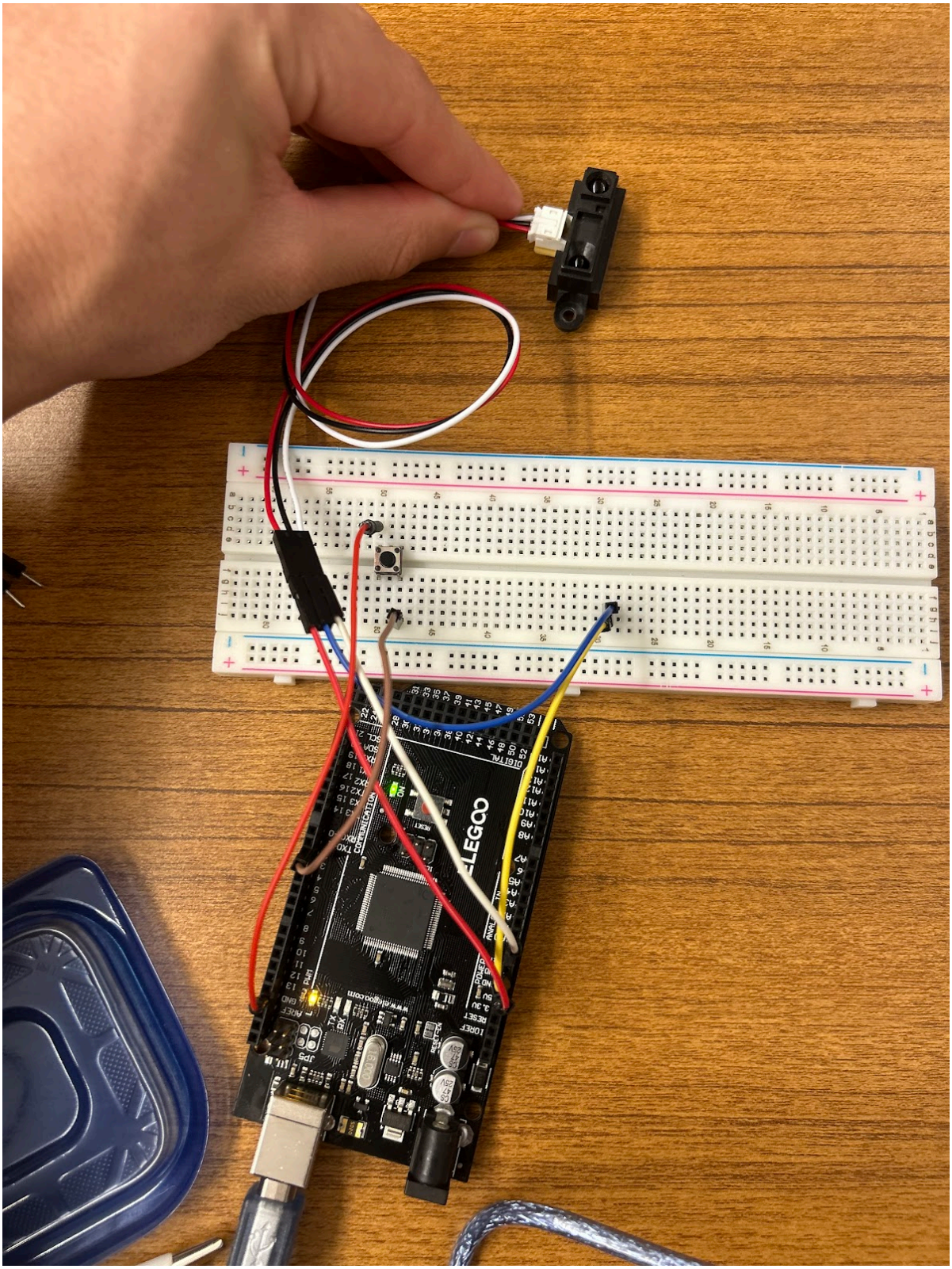
1. Mount the Sharp IR sensor horizontally on some type of test stand (emitter and receiver are side by side). The sensor should be at least 3 inches from the table.
2. Connect the infrared sensor power wires to the Arduino 5V and ground outputs. There is a special cable provided.
3. Measure the voltage at the output port of the sensor with a multimeter as you position the sensor at specific distances from your "wall." You can make a white wall by taping a piece of paper to a book or something similar or use a wall in the room. Record the voltage at 5, 10, 15, 20, 25, 30, 35, and 40 cm.
4. Graph the output voltage vs. distance for the measured points. (Hand draw or use Excel or equivalent) (You can do the graph later when you prepare the report if you want)



5. Connect the IR sensor output to one of the analog inputs of the Arduino.
6. Connect a pushbutton switch to one of the digital inputs to allow you to detect when the switch is pressed. Add a 100K pullup resistor like we learned in class or configure the input pin to use the internal pullup resistor.
7. Write a program that, **each time you push the button**:
   a) Measures the analog input voltage from the sensor.
   b) Uses the information you gathered in the prior experiment to calculate the distance to the wall. **Use a piecewise linear model to approximate the conversion function**.
   c) The program should print the distance to the wall on the serial console **once each time you press the pushbutton**. Report the distance to the nearest 1mm when possible.

DEMO #1 – show the TA that this program works.

Circuit



CODE

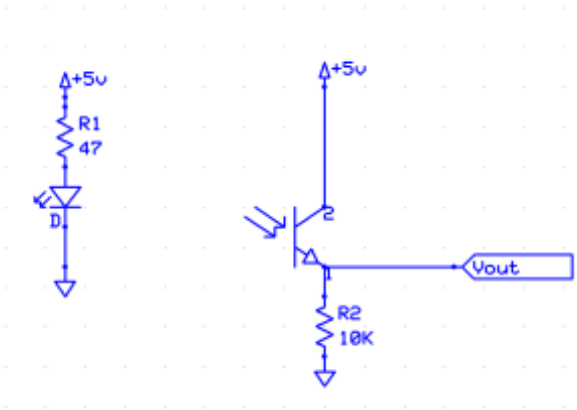
```
//demo1
#define PIN_NUM 2
#define ANALOG_PIN A0
int distances[8] = {50,100,150,200,250,300,350,400}; //mm
int voltages[8] = {3090, 2230, 1620, 1280, 1070, 912, 817, 710}; //mv
allows to avoid floats

void setup() {
  Serial.begin(9600);
  Serial.println("Setup complete.");
  pinMode(PIN_NUM, INPUT_PULLUP);
  attachInterrupt(digitalPinToInterrupt(PIN_NUM), isr0, FALLING);

  pinMode(ANALOG_PIN, INPUT); // Configure pin A0 as an input for analog
reading
}
volatile int flag = 0;
int millivolt = 0;
void loop() {
  if (flag) {
    int i;
    millivolt = (int)((long)analogRead(ANALOG_PIN) * 5000L/1024L);
    Serial.println("Millivolt is " + String(millivolt));
    for (i=7; i > 0; i--){
      if(millivolt >= voltages[i] && millivolt < voltages[i-1])
      {
        int calc_distance;
        float slope = ((float)voltages[i-1] - (float)voltages[i]) /
((float)distances[i-1] - (float)distances[i]);
        calc_distance = ((float)millivolt - (float)voltages[i]) / slope +
(float)distances[i];
        Serial.println("The voltage is " + String(millivolt) + "mV, and the
distance is " + String(calc_distance) + "mm.");
      }
    }
```

```
    flag = 0;
  }
}

void isr0() {
  flag = 1;
  Serial.println("we have been interrupted");
}
```

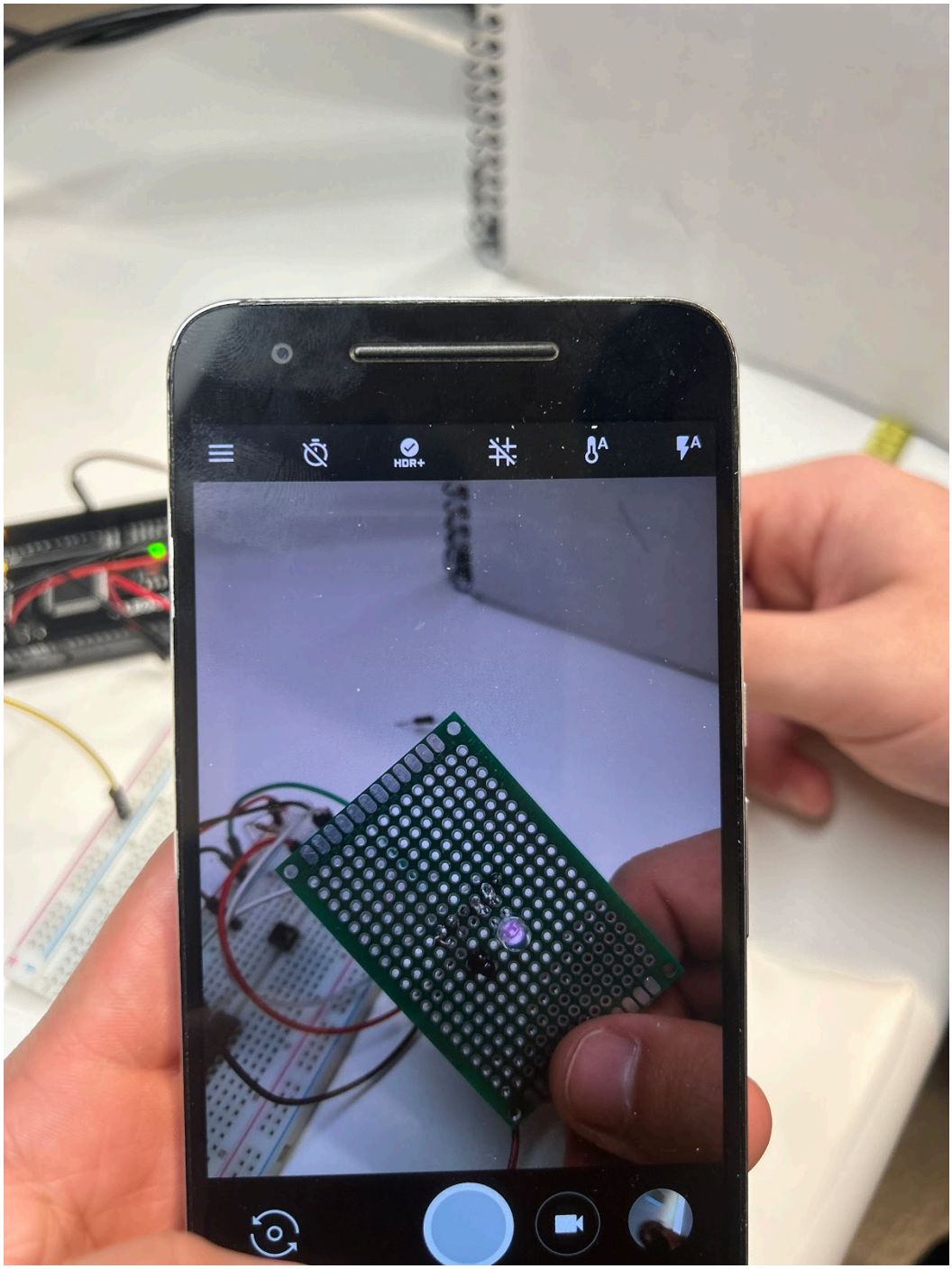
DEMO RESULTS
23cm: we have been interrupted Millivolt is 1142 The voltage is 1142mV, and the distance is 232mm.
10cm: we have been interrupted Millivolt is 2338 The voltage is 2338mV, and the distance is 93mm.

Part II: Build Your own sensor:

1. Next, you are going to build your own sensor. Follow the directions in the "Building an IR sensor" handout. **Review the LTE-5208A Infrared LED and the LTR-4206E Phototransistor documentation carefully to identify which pin is which**.
2. Connect the IR LED and Phototransistor to the following circuit.

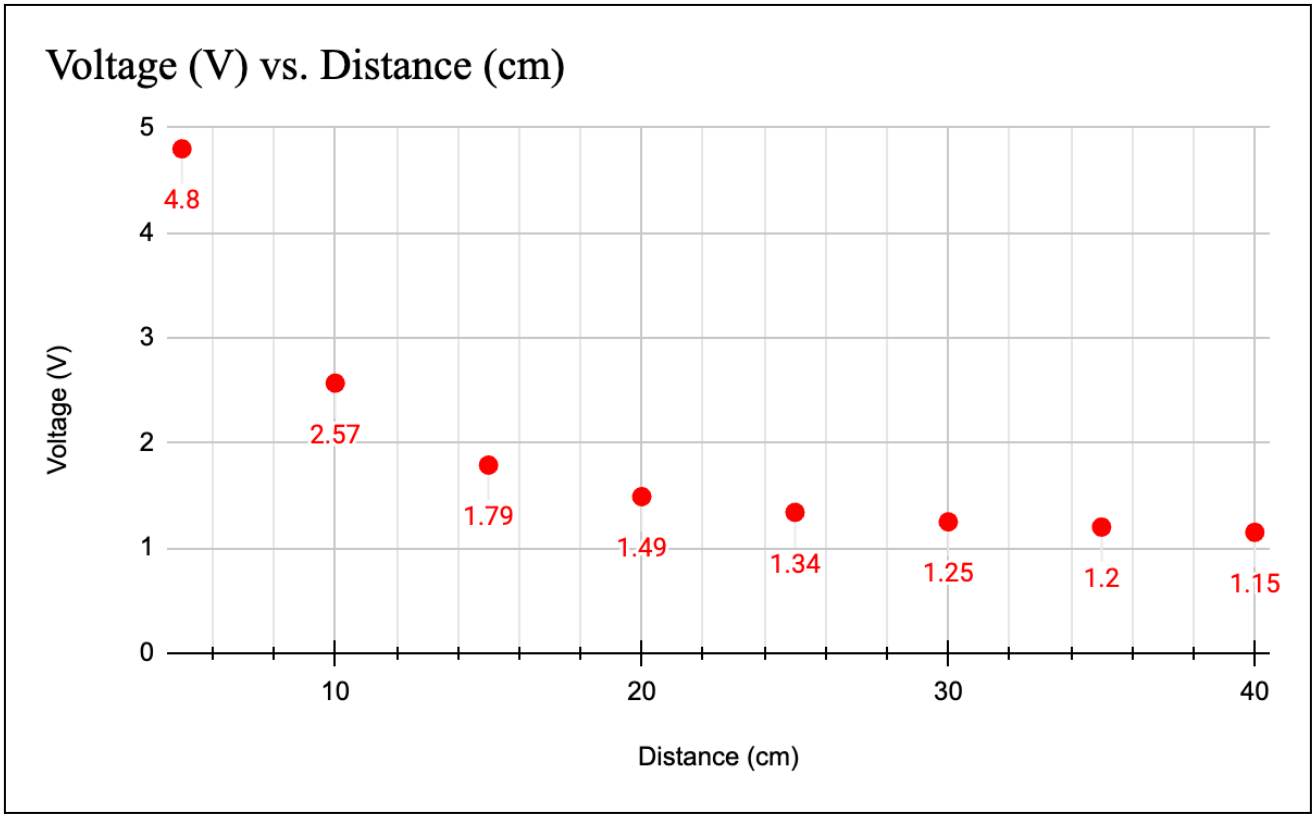


Measure the voltage drop across the LED terminals. Look at the LED through your phone camera (or mine)[1]. You may be able to see the glow. Report the voltage drop in your lab report and calculate the current through the LED. (Hint. Any time it says the word "measure" in a lab, the results belong in your report.)

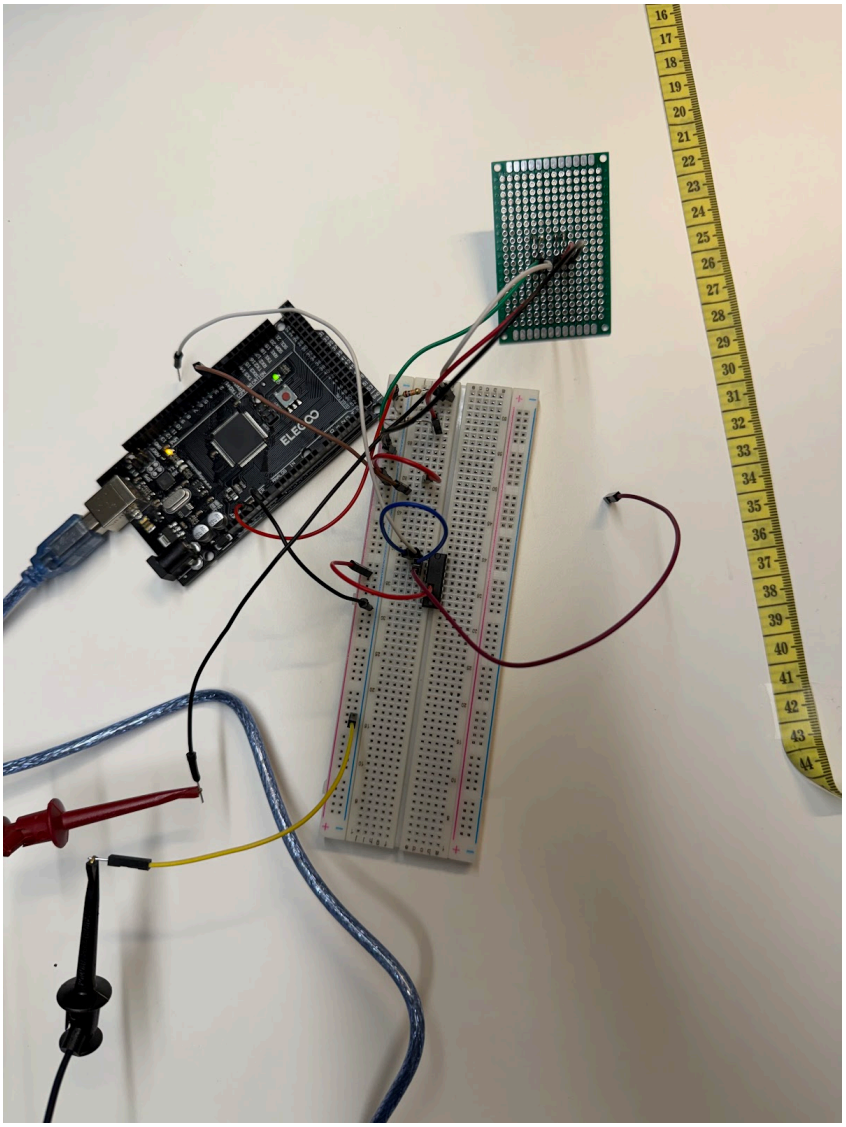
LED GLOWING:


Voltage drop across LED:1.348V
1.348/47 = 28.68mA

3. Measure the voltage at Vout with a multimeter as you position the sensor at specific distances from your "wall." Record the voltage at 5, 10, 15, 20, 25, 30, 35, and 40 cm. Graph in your report.

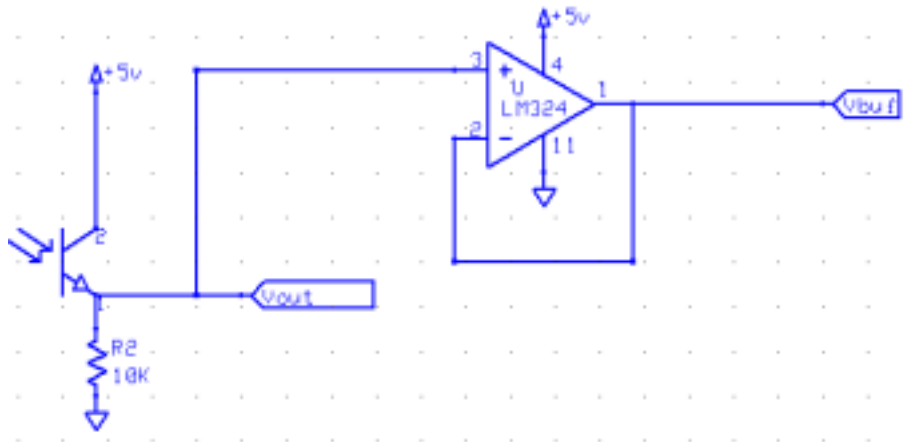


Circuit



DEMO #2 – show the TA that the voltage changes as the wall moves.

4. Add a voltage follower circuit using an LM324 op-amp. Measure Vbuf and Vout at 5 and 10cm. How do they compare to the prior measurements?



	VBUF	VOUT
5 cm	3.723V	4.702V
10 cm	2.370V	2.370V

We expect the Vbuf to be similar to Vout, but on the 5cm measurement, it didn't match. We put this down to some sort of railing discrepancy. Other, farther distances matched.

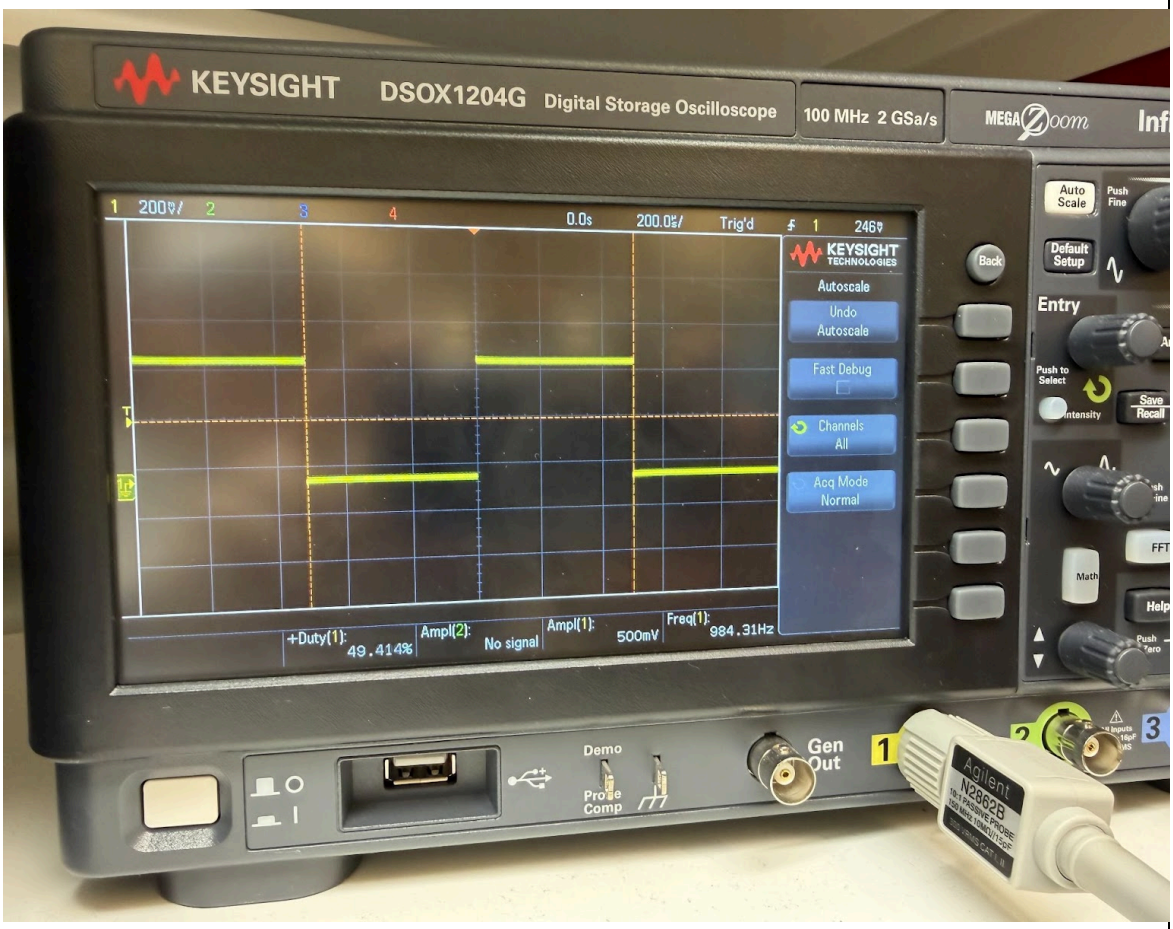
5. Now, write a program to generate a 1KHz square wave on pin 12 of the Arduino. Observe this signal on the Oscilloscope on channel 1. Record what you see on the scope. (Cell phone picture is easiest – but you can do a drawing if you want. You should generally record scope outputs for your report. If you bring a thumb drive, the scope can take screenshots)

CODE

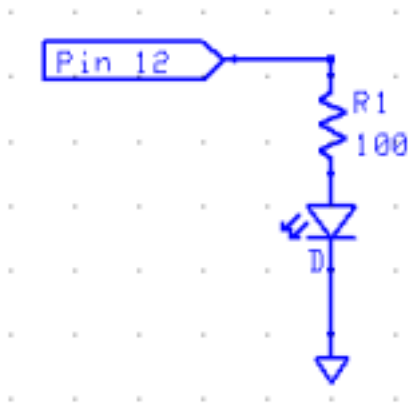
```
//demo3

unsigned long timeCheck;

void setup() {
  Serial.begin(9600);
  Serial.println("Setup complete.");
  pinMode(12, OUTPUT);
  timeCheck = micros();
}

void loop() {
```

```
  if (micros() - timeCheck >=500){
    if (micros() - timeCheck >=1000){
      digitalWrite(12, LOW);
      timeCheck = micros();
    }
    else digitalWrite(12, HIGH);
  }
}
```

Oscilloscope Square Wave



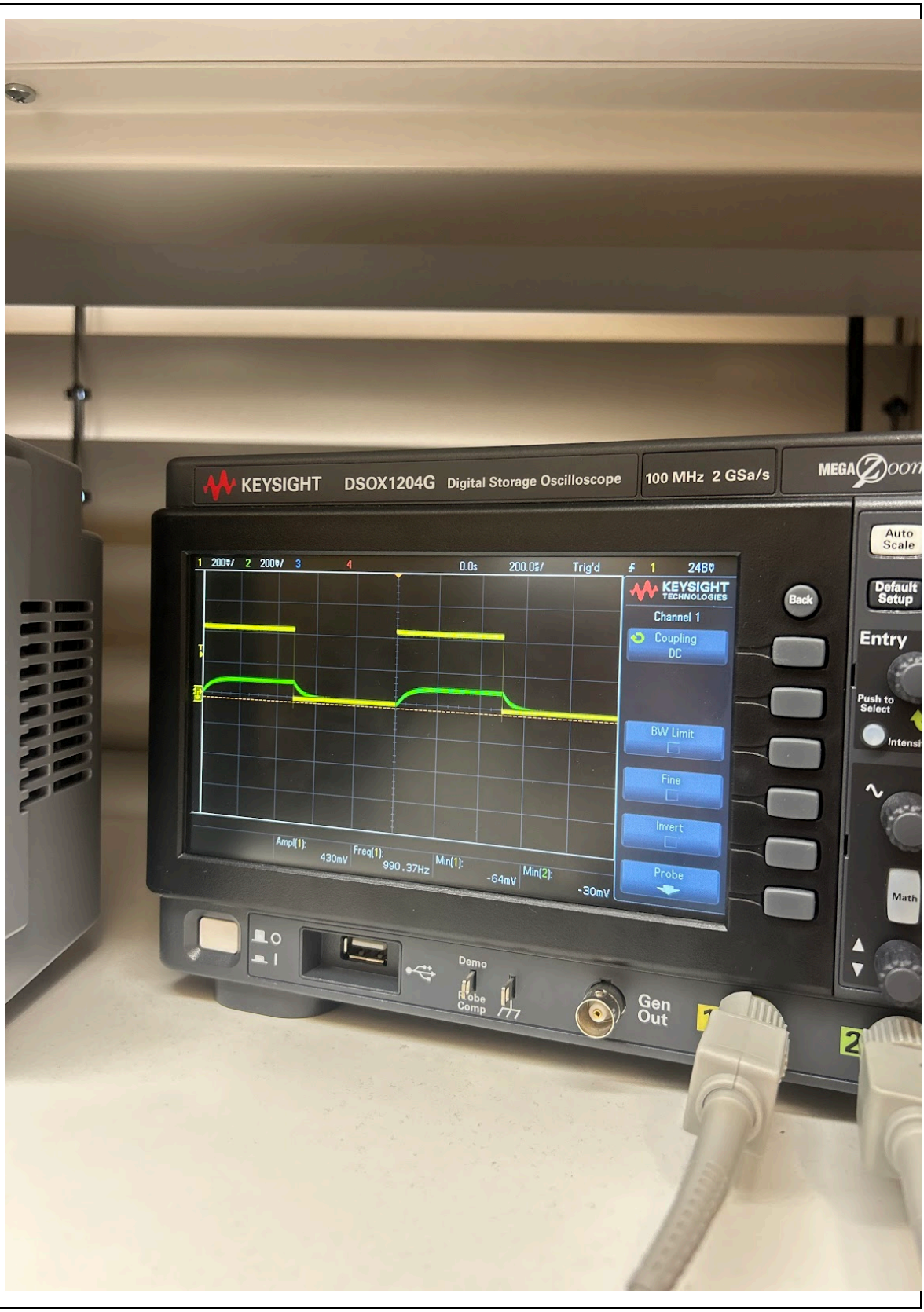
6. Change the LED circuit to look like this. Note the change in the resistor.

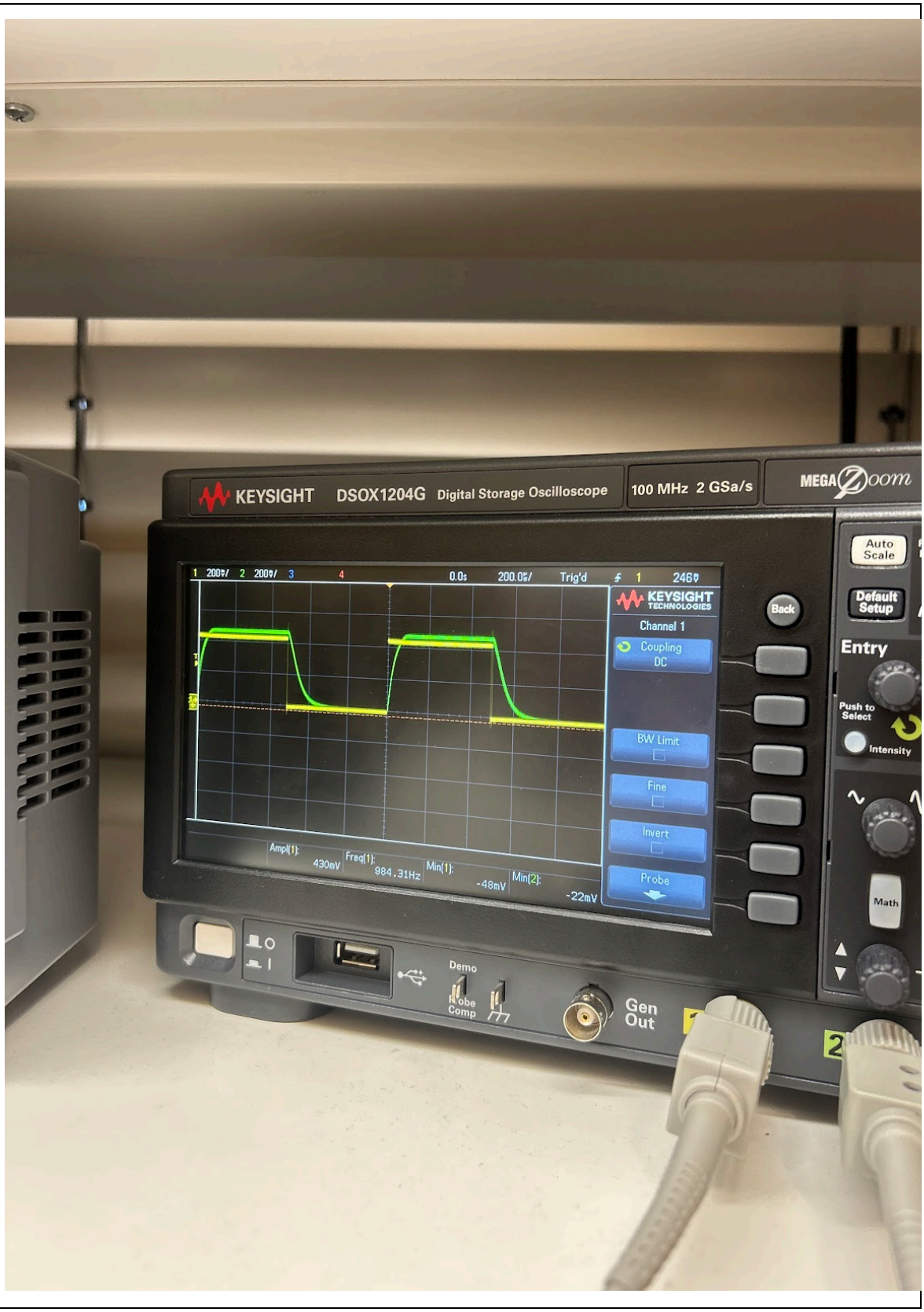


7. Leave Pin 12 on the Arduino connected to the oscilloscope channel 1. Connect Vbuf to channel 2. Trigger the scope off of channel 1. Record what you see. Move the wall various distances and observe what happens. DEMO #3 – show the TA the scope output

Oscilloscope Observation
Channel two's magnitude was dependent on the voltage across Vbuf which was expected. The closer the sensor was to the wall, the greater the magnitude.

Oscilloscope Pictures of Vbuf Compared to Input Wave


KEYSIGHT
DSOX1204G
Digital Storage Oscilloscope
100 MHz 2 GSa/s
MEGA Zoom
Auto Scale
Default Setup
Entry
Push to Select
Math
Back
KEYSIGHT TECHNOLOGIES
Channel 1
Coupling DC
BW Limit
Fine
Invert
Probe
Trig'd
Ampl(1): 430mV
Freq(1): 990.37Hz
Min(1): -64mV
Min(2): -30mV
Demo
Probe Comp
Gen Out


KEYSIGHT
DSOX1204G
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Entry
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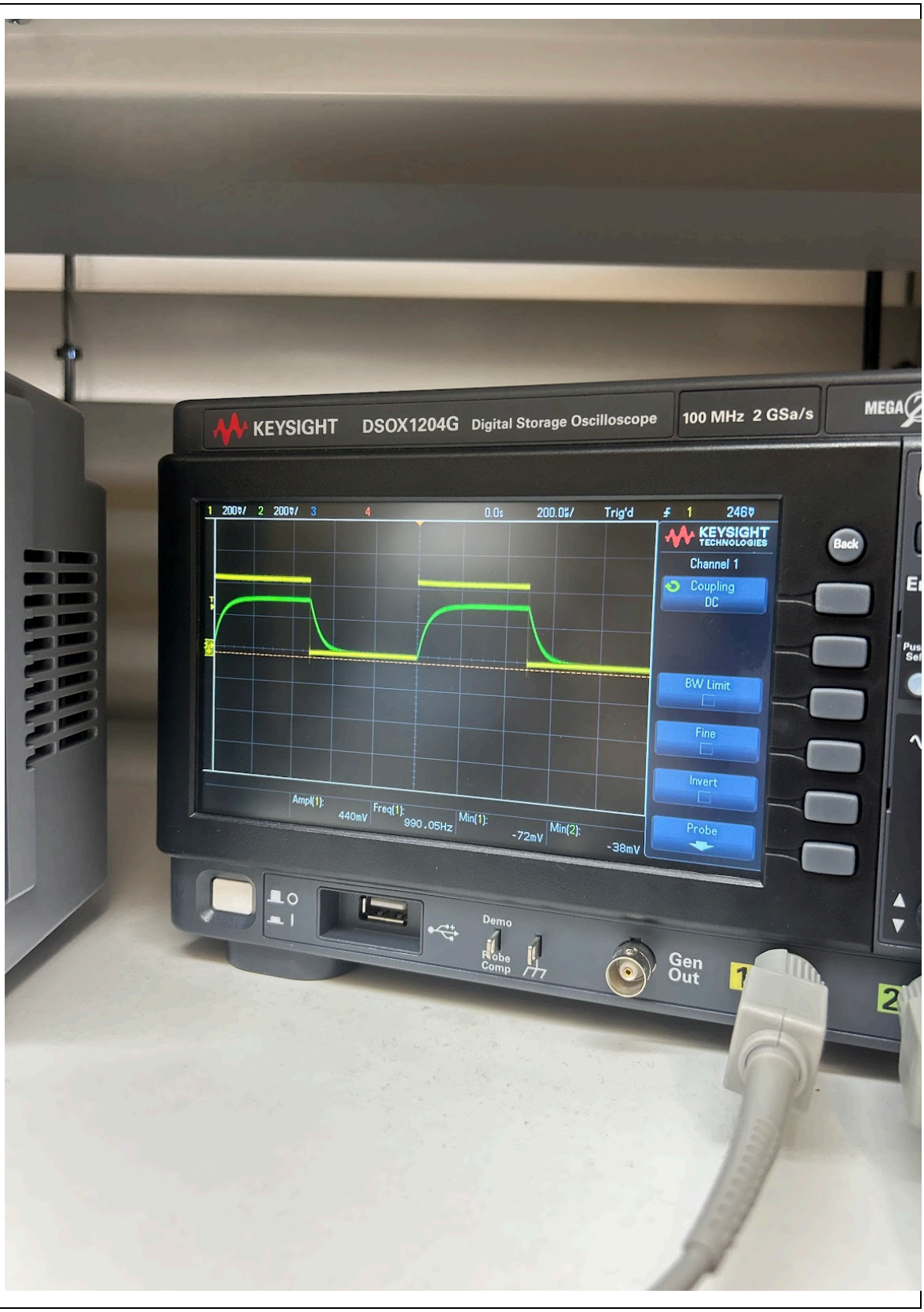
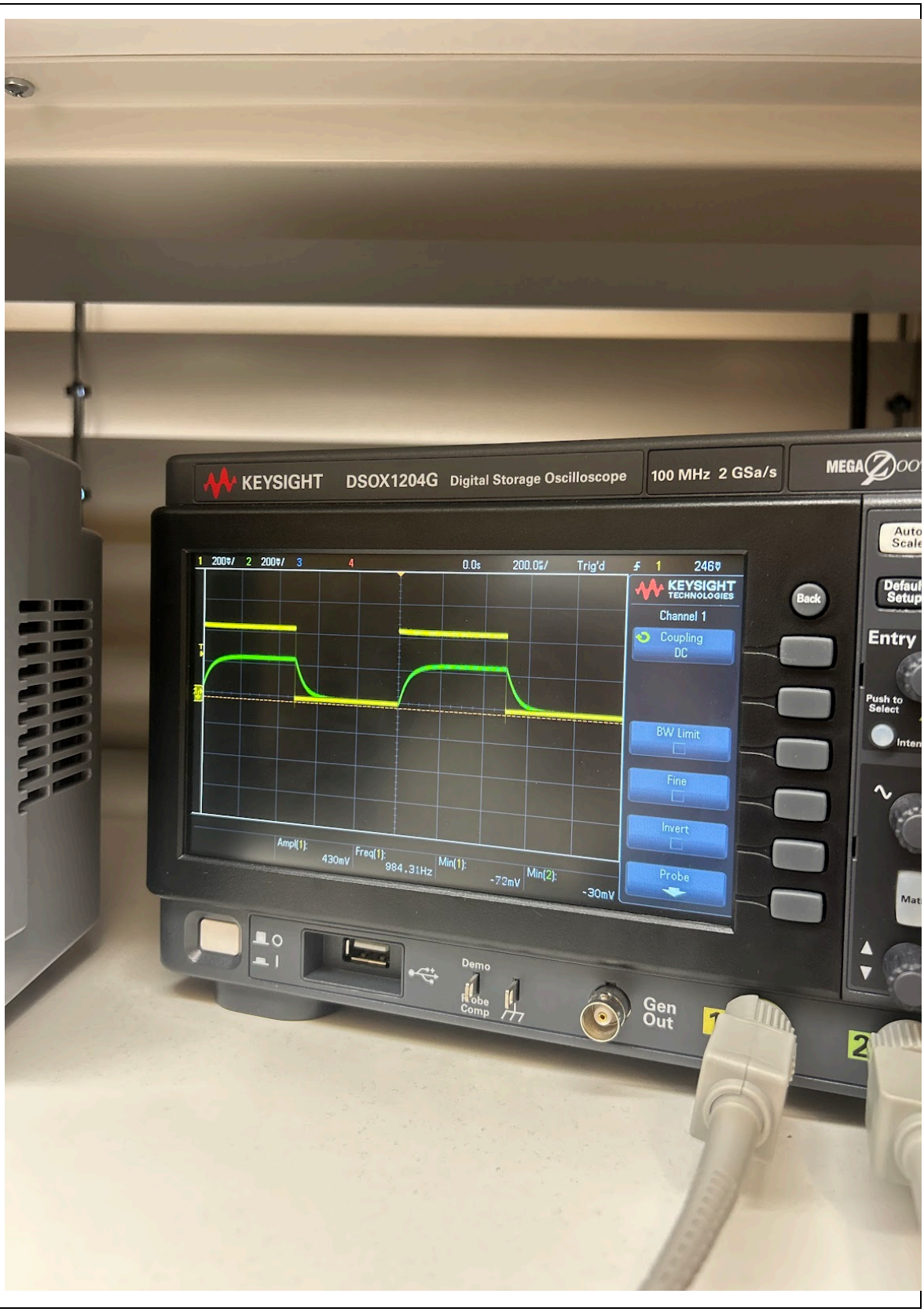
Intensity
Math
Back
KEYSIGHT TECHNOLOGIES
Channel 1
Coupling
DC
BW Limit
Fine
Invert
Probe
Trig'd
Ampl(1):
430mV
Freq(1):
984.31Hz
Min(1):
-48mV
Min(2):
-22mV
Demo
Probe Comp
Gen Out


KEYSIGHT
DSOX1204G Digital Storage Oscilloscope
100 MHz 2 GSa/s
Trig'd
KEYSIGHT TECHNOLOGIES
Channel 1
Coupling
DC
BW Limit
Fine
Invert
Probe
Back
Ampl(1):
440mV
Freq(1):
990.05Hz
Min(1):
-72mV
Min(2):
-38mV
Demo
Probe Comp
Gen Out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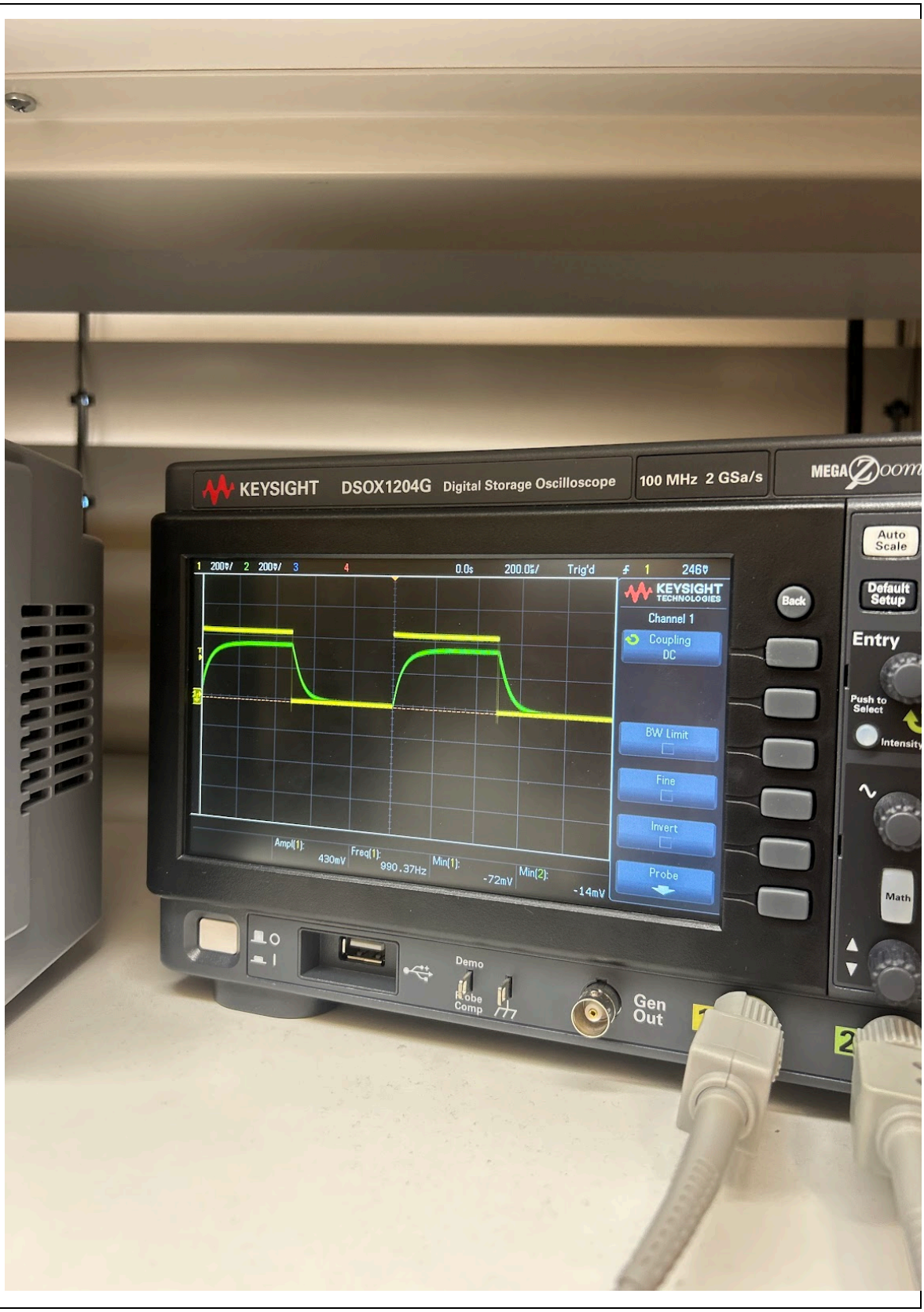
Freq(1):
984.31Hz
Min(1):
-72mV
Min(2):
-30mV
Trig'd
Entry
Push to Select
Demo
Probe Comp
Gen Out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Probe
Ampl(1):
430mV
Freq(1):
990.37Hz
Min(1):
-72mV
Min(2):
-14mV
Trig'd
Demo
Probe Comp
Gen Out